Now we can estimate,

$$\begin{aligned}
\|u - \mathcal{I}_h u\|_{m,\mathcal{T}_h} &= \left(\sum_{s=0}^{m} |u - \mathcal{I}_h u|^2_{s,\mathcal{T}_h}\right)^{\frac{1}{2}} \\
&\leq C\left(\sum_{s=0}^{m} h^{4-2s}|u|^2_{2,\Omega}\right)^{\frac{1}{2}} \\
&\leq Ch^{2-m}|u|_{2,\Omega}
\end{aligned}$$

where we have only kept the leading order error term. ■

Note that the preceding theorem shows in particular

$$\|u - \mathcal{I}_h u\|_{1,\Omega} \leq ch|u|_{2,\Omega}$$

As anticipated in eq. (13.2), we can connect this estimate to the optimality result derived in section 12.2. We thus obtain the main result for the error of the Ritz-Galerkin approximation as

$$\|u - u_h\|_{1,\Omega} \leq ch|u|_{2,\Omega}.$$

Also note that, while we have an interpolation error of $\mathcal{O}(h^2)$ for the L_2-norm (i.e. for $m = 0$), we cannot trivially infer an error of this order for the Ritz-Galerkin approximation, because we have nothing to which we can connect this result. The optimality result for the H^1 norm follows from the fact the we could show boundedness and coercivity in that norm. We have no corresponding result for the L_2 norm. Nevertheless, we would hope that we gain an order of magnitude in convergence when measured in L_2 norm, just like for the interpolation error. This is the topic of the next section.

13.7 L_2 Estimates

Although the analysis of the interpolation error yields an estimate in L_2, we have no optimality result, i.e. the equivalent of theorem 12.2, in that norm to which we could connect it. Nevertheless, we show in this section that, under mild assumptions, one can get error estimates also in L_2.

As an example, consider the variational formulation of the elliptic model problem with homogeneous boundary conditions (cf. chapter 11). We can state this as the problem of finding $u \in H^1_0(\Omega)$, such that

$$a(u, v) = l(v), \quad \forall v \in H^1_0(\Omega) \tag{13.17}$$

where $a(u, v) = \int_\Omega (\nabla v)^T A(x)\nabla u \, dx$ and $l(v) = (f, v)_{0,\Omega}$. (See the definition of Model Problem 4 for definition of the data.) We have already shown that this problem admits a unique solution, and we assume that Theorem 11.5 holds. This means $u \in H^2$ and for some positive constant C, we have

$$\|u\|_{2,\Omega} \leq C \|f\|_{0,\Omega}. \tag{13.18}$$

Let the solution of the Ritz-Galerkin approximation to (13.17), based on the linear elements introduced in Section 13.3, be denoted as u_h. We would like to bound the L_2 error $\|u - u_h\|_{0,\Omega}$. The main tool is to consider the "dual" problem of finding $w \in H^1_0(\Omega)$, such that

$$a(v, w) = (u - u_h, v), \quad \forall v \in H^1_0(\Omega) \tag{13.19}$$

The data to the dual problem is thus given by the error of the Ritz-Galerkin approximation to (13.17). Recall that $u \in H^2(\Omega) \subset L_2(\Omega)$, and by construction, $u_h \in S_h \subset H^1_0(\Omega) \subset L_2(\Omega)$, and the data of the dual problem is thus in L_2. Since $a(\cdot, \cdot)$ is symmetric, the dual problem is

the same as the original problem with a different right-hand side.[6] In analogy to (13.18), we have for the solution

$$\|w\|_{2,\Omega} \leq C \|u - u_h\|_{0,\Omega} . \tag{13.20}$$

Using this, we can derive an estimate for the L_2-error $\|u - u_h\|_{0,\Omega}$ of the Ritz-Galerkin approximation to (13.17):

Theorem 13.5 *Let u be the solution to problem* (13.17)*, and let u_h be the corresponding Ritz-Galerkin approximation based on the linear elements introduced in 13.3. Assume the domain and the data of the problem are such that* (13.18) *holds. Then*

$$\|u - u_h\|_{0,\Omega} \leq ch^2 |u|_{2,\Omega}$$

Proof For $u = u_h$ the proof is trivial. Assume in the following $u \neq u_h$, and consider the dual problem (13.19). Set $v = u - u_h$ to obtain

$$a(u - u_h, w) = (u - u_h, u - u_h) = \|u - u_h\|_{0,\Omega}^2$$

Recall that the bilinear form corresponding to the Poisson problem is continuous, and use Galerkin orthogonality $a(u - u_h, \mathcal{I}_h w) = 0$, where $\mathcal{I}_h$ is the interpolation operator introduced in Section 13.3, to obtain

$$\begin{aligned} \|u - u_h\|_{0,\Omega}^2 = a(u - u_h, w - \mathcal{I}_h w) &\leq C_1 \|u - u_h\|_{1,\Omega} \|w - \mathcal{I}_h w\|_{1,\Omega} \\ &\leq C_2 h \|u - u_h\|_{1,\Omega} |w|_{2,\Omega} \\ &\leq C_2 h \|u - u_h\|_{1,\Omega} \|w\|_{2,\Omega} \\ &\leq C_3 h \|u - u_h\|_{1,\Omega} \|u - u_h\|_{0,\Omega} \end{aligned}$$

where we have used the interpolation error derived in Theorem 13.4, and eq. (13.20). Dividing by the common factor $\|u - u_h\|_{0,\Omega}$, and using the already available H^1 error estimate, we have

$$\|u - u_h\|_{0,\Omega} \leq \tilde{c}h \|u - u_h\|_{1,\Omega} \leq ch^2 |u|_{2,\Omega}$$

■

13.8 Further Reading

The derivation of the interpolation error is very similar to the exposition in Braes [3]. An alternative presentation is given in Brenner [4].

[6] Note that for the dual problem the unknown function w is the second input argument of the bilinear form $a(\cdot, \cdot)$, while the test function v is the first argument. For a symmetric form, this makes no difference, but we keep this notation anyway, as it is compatible with the dual problem for the more general non-symmetric case.

Chapter 14

Extensions to General Elliptic Problems

14.1 Inhomogeneous Boundary Conditions

Consider again Model Problem 4, and assume, for simplicity, that the matrix A in eq. (11.1) is the identity matrix. Instead of setting $u = 0$ at the boundary, we now prescribe a function g:

$$\begin{aligned} -\nabla^2 u &= f, \qquad x \in \Omega, \\ u &= g, \qquad x \in \partial\Omega. \end{aligned}$$

For simplicity, we assume that both f and g are smooth functions. Multiply the PDE with a test function $v \in V_0$, where V_0 needs to be specified, and integrate by parts to obtain

$$\int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \nu \cdot \nabla u \, v ds = \int_\Omega f v dx,$$

where ν is the outward pointing normal on the boundary $\partial\Omega$. Note that the second term on the left-hand side destroys the symmetry of the problem. We can eliminate it by setting $V_0 \equiv H_0^1(\Omega)$, as for the homogeneous problem. Note, however, that the boundary conditions do not appear in the weak formulation. It makes sense, therefore, to incorporate boundary conditions into the solution space. That is, we seek $u \in V_D$, where

$$V_D := \{v \in H^1(\Omega) : tr\ v = g\},$$

where $tr\ v = g$ is the restriction of v to the boundary.[1] The problem is thus to find $u \in V_D$ such that

$$a(u, v) = l(v), \quad \forall v \in V_0. \tag{14.1}$$

where

$$\begin{aligned} a(u, v) &= \int_\Omega \nabla u \cdot \nabla v \, dx, \\ l(v) &= \int_\Omega f v dx. \end{aligned}$$

[1] In general, it is problematic to think of the trace $tr\ v$ merely as the restriction $v|_{\partial\Omega}$ of a function to the boundary. Recall that in our weak formulation data on sets of measure zero in $\mathbb{R}^d$ is not well defined. The trace should be thought of as a mapping from some functional space defined in the interior of the domain into some functional space on the boundary $\partial\Omega \subset \mathbb{R}^{d-1}$. However, the correct exposition of this matter is somewhat technical. Under our present assumption of smooth data, we can recover the strong solution from the weak formulation where the restriction of functions to the boundary is well-defined in the traditional sense. We use this as an excuse to avoid dealing with these more subtle issues, and simply think of the trace as the restriction operator.

We cannot directly use Theorem 11.4 to show that the problem is well posed, because solution space and test space are not the same. (Furthermore, V_D is not a linear space to begin with. For example, for $v, w \in V_{h,D}$, we have $v + w \notin V_{h,D}$!) Nevertheless, we can show that the problem is well posed by a slight re-formulation. But first we note that any solution must be unique. Indeed, if u_1, u_2 are two solutions to (14.1), the function $w := u_1 - u_2 \in V_0$ solves $a(w, v) = 0$ for all $v \in V_0$ which admits only the trivial solution, and hence $u_1 = u_2$.

For the finite-element formulation we may define the discrete approximation spaces in a manner similar to that of Section 13.3, as the continuous, piecewise linear approximation that satisfies the boundary conditions required in the respective spaces. Let all the assumptions regarding the mesh and approximation space made in Chapter 13 hold. We may use the interpolation operator $\mathcal{I}_h$ defined in Section 13.3 to define the finite-dimensional spaces:

$$\begin{aligned} V_{h,0} &:= \{\mathcal{I}_h v : v \in V_0\} \\ V_{h,D} &:= \{\mathcal{I}_h v : v \in V_D\}. \end{aligned} \tag{14.2}$$

Now the discrete problem becomes to find $u_h \in V_{h,D}$, such that

$$a(u_h, v_h) = l(v_h), \quad \forall v \in V_{h,0}. \tag{14.3}$$

We now show that (14.1) and (14.3) are well posed. Consider arbitrary $\widetilde{u} \in V_D$ (which implies $tr\ \widetilde{u} = g$), and the following problem: Find $z \in V_0\ (= H_0^1(\Omega))$, such that

$$a(z, v) = \underbrace{l(v) - a(\widetilde{u}, v)}_{=:\widetilde{l}(v)}, \quad \forall v \in V_0.$$

To this new problem we can apply Theorem 11.4, and we find that the problem has a unique solution. (See Example 11.1 for continuity and coercivity of the bilinear form $a(\cdot, \cdot)$, and the continuity of the linear functional l. Using the continuity of the bilinear form, it is easy to see that the modified functional $\widetilde{l}$ is still continuous.) The function $u = z + \widetilde{u}$ solves (14.1), which thus also possesses a solution. (And that solution must be unique, as we have already seen.)

Now we can consider $\widetilde{u}_h \in V_{h,D}$, and solve for $z_h \in V_{h,0} \subset V_0$, by setting

$$a(z_h, v_h) = \underbrace{l(v_h) - a(\widetilde{u}_h, v_h)}_{=:\widetilde{l}(v_h)}, \quad \forall v_h \in V_{h,0}.$$

Repeating the arguments applied to the infinite-dimensional problem, we see that this problem has a unique solution and is equivalent to (14.3). We obtain $u_h = z_h + \widetilde{u}_h$.

However, it would be inconvenient for both the implementation of the problem, and for further analysis, if we had to explicitly introduce a new function $\widetilde{u}$ into the problem. Fortunately we do not have to do this for either. The re-formulation helped us to show that the problem admits a solution. As far as implementation goes, it is clear that one can implement in the original formulation (14.3), which is, after all, equivalent. How to proceed with the error analysis, however, is again not so clear. For example, we need again coercivity. If $a(\cdot, \cdot)$ takes input arguments from two different spaces (without one being a subset of the other), it is not so clear how to define coercivity. It turns out, however, that for the error analysis it will be sufficient to require coercivity on $V_{h,0}$. This allows us to recover Theorem 12.2. We again concentrate on problems formulated in H^1, i.e. $V_0 \equiv H_0^1(\Omega)$, so the finite-dimensional setting which we have used so far, applies.

Theorem 14.1 *Let $u \in V_D$ solve the variational problem* (14.1)*, and let $u_h \in V_{h,D}$ solve* (14.3)*. Assume that there is a constant $\alpha > 0$,*

$$a(v_h, v_h) \geq \alpha \|v_h\|_{1,\Omega}^2, \quad \forall v_h \in V_{h,0}, \tag{14.4}$$

where $V_{h,0} \subset V_0 \equiv H_0^1(\Omega)$ is defined in (14.2)*, and, for some $C > 0$,*

$$a(u,v) \le C \|u\|_{1,\Omega} \|v\|_{1,\Omega}, \quad \forall u, v \in H^1(\Omega). \tag{14.5}$$

Then

$$\|u - u_h\|_{1,\Omega} \le \left(1 + \frac{C}{\alpha}\right) \inf_{v \in V_{h,D}} \|u - v\|_{1,\Omega} \tag{14.6}$$

PROOF Let $v \in V_{h,D}$ be arbitrary. By the triangle inequality, we have

$$\|u - u_h\|_{1,\Omega} = \|u - v + v - u_h\|_{1,\Omega} \le \|u - v\|_{1,\Omega} + \|v - u_h\|_{1,\Omega}.$$

We need to re-work the second term. Note that $v - u_h \in V_{h,0}$, so we can use coercivity on $V_{h,0}$, i.e. inequality (14.4), to obtain

$$\begin{aligned} \alpha \|v - u_h\|^2 &\le a(v - u_h, v - u_h) \\ &= a(v - u, v - u_h) + a(u - u_h, v - u_h). \end{aligned} \tag{14.7}$$

Noting that $V_{0,h} \subset V_0$, we see from (14.1) and (14.3) that we satisfy Galerkin orthogonality, i.e.

$$a(u - u_h, v_h) = 0, \quad \forall v_h \in V_{h,0}.$$

Consequently, the last term in (14.7) vanishes. Using continuity (14.5), we have

$$\|v - u_h\|_{1,\Omega}^2 \le \frac{C}{\alpha} \|v - u\|_{1,\Omega} \|v - u_h\|_{1,\Omega},$$

and consequently

$$\|u - u_h\|_{1,\Omega} \le \left(1 + \frac{C}{\alpha}\right) \|u - v\|_{1,\Omega} \qquad \forall v \in V_{h,D}.$$

The assertion follows. ■

As a simple Corollary, using the interpolation result proved in Theorem 13.4 (which still hold for non-zero boundary conditions, as long g is smooth enough.), we have

$$\|u - u_h\|_{1,\Omega} \le \gamma h |u|_{2,\Omega}. \tag{14.8}$$

14.2 General Elliptic Problems

We now consider a more general elliptic problem on a bounded domain Ω. First, we write the general d-dimensional, second-order elliptic PDE

$$-\sum_{i,j=1}^{d} (a_{ij}(x) u_{x_j})_{x_i} + \sum_{i=1}^{d} (b_i(x) u)_{x_i} + c(x) u = f(x), \qquad x \in \Omega, \tag{14.9}$$

where $a_{ij} \in L_\infty$, and the matrix $(a_{ij})_{i,j=1,\ldots,d}$ is symmetric. Furthermore, we recall Definition 10.1, i.e. the definition of uniform ellipticity: For all $\xi \in \mathbb{R}^d$, and all $x \in \Omega$, there holds for some constant $\alpha' > 0$,

$$\sum_{i,j=1}^{d} \xi_i a_{ij}(x) \xi_j \ge \alpha' |\xi|^2.$$

We assume $f \in L_2(\Omega)$, and we let $c \in L_\infty(\Omega)$ be bounded away from zero, i.e. there is a constant $c_0 > 0$, such that $c(x) \geq c_0 > 0$. (We should state at this point that some of the assumptions we make in this section are not necessarily because the problem would otherwise not have a solution. In some cases we make assumptions to simplify the analysis a bit. We comment on this point later.) We also consider general boundary conditions. Assume that the boundary $\partial\Omega$ is partitioned into non-overlapping segments $\partial\Omega_D$ and $\partial\Omega_R$, such that

$$\begin{aligned} u &= g_D, \quad x \in \partial\Omega_D \\ \sum_{i,j=1}^{d} \nu_i a_{ij} u_{x_j} + \gamma u &= g_R, \quad x \in \partial\Omega_R \end{aligned}$$

where ν is the outward pointing normal on the boundary $\partial\Omega$, and γ, g_D, and g_R are smooth bounded functions. We assume that $\gamma(x) \geq 0$, for all $x \in \partial\Omega_R$. We designate these boundary conditions as (in that order) Dirichlet and Robin boundary conditions. (This should be compared to the terminology briefly discussed in Chapter 2, when we introduced the heat equation. Note that, in particular we can have $\gamma \equiv 0$, in which case we use the designation *natural* boundary conditions, instead of Robin boundary conditions.)

Defining $A := (a_{ij})_{i,j=1,\ldots d}$ and $\mathbf{b} = (b_1, \ldots, b_d)^T$, the problem can be written in concise form as

$$\begin{aligned} -\nabla \cdot (A\nabla u) + \nabla \cdot (\mathbf{b}u) + cu &= f, \qquad x \in \Omega. \\ u &= g_D, \quad x \in \partial\Omega_D \\ \nu \cdot (A\nabla u) + \gamma u &= g_R, \quad x \in \partial\Omega_R. \end{aligned}$$

In this section, we assume that $\mathbf{b} \equiv 0$. Problems with non-zero vectors $\mathbf{b}$ will be dealt with later. The weak formulation can be obtained, as usual, by multiplying with a smooth function and integrating by parts,

$$\begin{aligned} -\int_\Omega (\nabla \cdot (A\nabla u) + cu)\, v\, dx &= \int_\Omega f v\, dx \\ \iff \quad \int_\Omega (\nabla v)^T A \nabla u\, dx - \int_{\partial\Omega} \nu \cdot (A\nabla u) v\, dx + \int_\Omega cuv\, dx &= \int_\Omega f v\, dx \end{aligned}$$

The second term is non-symmetric. However, if we take v to vanish on $\partial\Omega_D$, and use the boundary conditions on $\partial\Omega_R$, we have

$$\int_\Omega (\nabla v)^T A \nabla u\, dx + \int_{\partial\Omega_R} \gamma uv\, dx + \int_\Omega cuv\, dx = \int_\Omega f v\, dx + \int_{\partial\Omega_R} g_R v\, dx.$$

We may formulate the problem in an abstract manner as the task of finding $u \in V_D$, such that

$$a(u, v) = l(v), \qquad \forall v \in V_{0,D}, \tag{14.10}$$

where

$$\begin{aligned} V_D &:= \{v \in H^1(\Omega) : \ \operatorname{tr} v = g_D, \ x \in \partial\Omega_D\}, \\ V_{0,D} &:= \{v \in H^1(\Omega) : \ \operatorname{tr} v = 0, \ x \in \partial\Omega_D\}, \end{aligned}$$

and

$$a(u, v) = \int_\Omega (\nabla v)^T A \nabla u\, dx + \int_{\partial\Omega_R} \gamma uv\, dx + \int_\Omega cuv\, dx, \tag{14.11}$$

$$l(v) = \int_\Omega f v\, dx + \int_{\partial\Omega_R} g_R v\, dx. \tag{14.12}$$

To show that this problem has a unique solution, we use an approach similar to that which has been used when first introducing inhomogeneous boundary conditions. For some $\widetilde{u} \in V_D$. we solve the problem of finding $z \in V_{0,D}$, such that

$$a(z,v) = \underbrace{l(v) - a(\widetilde{u},v)}_{=:\widetilde{l}(v)}, \qquad \forall v \in V_{0,D}. \tag{14.13}$$

This problem can be dealt with using our usual well-posedness theory, and repeating the arguments of section 14.1, we note that this formulation is equivalent to the original problem. We must show that the bilinear form is coercive (on $V_{0,D}$) and bounded, and $\widetilde{l}$ is bounded also. Since now more terms appear in the bilinear form this is not trivial. Note, however, the following theorem:

Theorem 14.2 *Let $a(\cdot,\cdot)$ be the bilinear form defined in* (14.11)*, and let l be the linear functional defined in* (14.12)*. Assume the functions appearing in these definitions satisfy the properties stated in section 14.2. Then problem* (14.10) *has a unique solution.*

Proof All we need to show that $a(u,v)$ is coercive on $V_{0,D}$, and furthermore, that $a(u,v)$ and $\widetilde{l}(v)$ (the latter is defined in (14.13)) are bounded. We start by showing coerciveness. For any $v \in H^1(\Omega)$ we have

$$\begin{aligned} a(v,v) &= \int_\Omega (\nabla v)^T A \nabla v \, dx + \int_{\partial\Omega_R} \gamma v^2 \, dx + \int_\Omega c v^2 \, dx \\ &\geq \alpha' \int_\Omega |\nabla v|^2 \, dx + \int_{\partial\Omega_R} \gamma v^2 \, dx + \int_\Omega c v^2 \, dx \\ &\geq \alpha' \|\nabla v\|_{0,\Omega}^2 + c_0 \|v\|_{0,\Omega}^2 \\ &\geq \alpha \|v\|_{1,\Omega}^2 \end{aligned} \tag{14.14}$$

where $\alpha = \min(\alpha', c_0)$. The bilinear form is thus coercive on all of $H^1(\Omega)$. Note that we have used the inequality $\gamma \geq 0$ which includes in particular the possibility that γ vanishes. In this simple analysis, we cannot allow $c(x)$ to vanish as well. For otherwise, we cannot bound α away from zero. See, however, Remark 14.1 at the end of this section for a way to show coerciveness on $V_{0,D}$ for the case that $c(x)$ is *not* bounded away from zero. Now we show continuity:

$$\begin{aligned} a(u,v) &= \int_\Omega (\nabla v)^T A \nabla u \, dx + \int_{\partial\Omega_R} \gamma u v \, dx + \int_\Omega c u v \, dx \\ &\leq \max_{i,j} \sup_{x\in\Omega} |a_{ij}(x)| \int_\Omega (\nabla v)^T \nabla u \, dx + \sup_{x\in\Omega_R} \{\gamma(x)\} \int_{\partial\Omega_R} u v \, dx + \sup_{x\in\Omega} \{c(x)\} \int_\Omega u v \, dx \\ &\leq \max_{i,j} \sup_{x\in\Omega} |a_{ij}(x)| \, \|\nabla v\|_{0,\Omega} \|\nabla u\|_{0,\Omega} + \sup_{x\in\partial\Omega_R} \{\gamma(x)\} \|u\|_{0,\partial\Omega} \|v\|_{0,\partial\Omega} + \sup_{x\in\Omega} \{c(x)\} \|u\|_{0,\Omega} \|v\|_{0,\Omega} \end{aligned}$$

Where the Cauchy-Schwarz inequality has been used in the last inequality. We also have the trace inequality $\|v\|_{0,\partial\Omega_R}^2 \leq c \|v\|_{1,\Omega}^2$ for some constant $c > 0$ (We omit proof of this last inequality here.) Then we can conclude

$$\begin{aligned} a(u,v) &\leq \max_{i,j} \sup_{x\in\Omega} |a_{ij}(x)| \, \|\nabla v\|_{0,\Omega} \|\nabla u\|_{0,\Omega} + c^2 \sup_{x\in\Omega_R} \{\gamma(x)\} \|u\|_{1,\Omega} \|v\|_{1,\Omega} + \sup_{x\in\Omega} \{c(x)\} \|u\|_{0,\Omega} \|v\|_{0,\Omega} \\ &\leq C \|u\|_{1,\Omega} \|v\|_{1,\Omega} \end{aligned}$$

using the obvious inequality $\|\cdot\|_{0,\Omega} \leq \|\cdot\|_{1,\Omega}$, and $|\cdot|_{1,\Omega} \leq \|\cdot\|_{1,\Omega}$. Similarly

$$\begin{aligned} |l(v)| &\leq \left| \int_\Omega f v \, dx \right| + \left| \int_{\partial\Omega_R} g_R v \, dx \right|, \\ &\leq \|f\|_{0,\Omega} \|v\|_{0,\Omega} + \|g_R\|_{0,\partial\Omega_R} \|v\|_{0,\partial\Omega_R} \end{aligned} \tag{14.15}$$

Again using the trace inequality, and the trivial inequality $\|v\|_{0,\Omega} \leq \|v\|_{1,\Omega}$, we conclude

$$|l(v)| \leq C \|v\|_{1,\Omega} \tag{14.16}$$

Note that we have assumed g_R to be smooth and bounded on $\partial\Omega_R$, so it is certainly square integrable. From eq. (14.15) it seems that assuming $g_R \in L^2(\partial\Omega)$ is enough, but not every function in $L^2(\partial\Omega)$ is the trace of some function $\widetilde{u} \in H^1(\Omega)$. We again refer to Remark 14.1 at the end of this section.

Together with the boundedness of a eq. (14.16) implies the boundedness of $\widetilde{l}$. ■

The error analysis proceeds in a manner similar to Theorem 14.1, and shall be omitted.

Remark 14.1 As stated previously in this section, some of the assumptions regarding the coefficient functions, and the boundery conditions are made primarily to simplify the analysis, not because otherwise we could not show that there is a unique solution. An example is given by the function $c(x)$. It is convenient to assume that $c(x) \geq c_0 > 0$, as this allows us to conclude that $a(\cdot,\cdot)$ is coercive very easily (cf. (14.14)). It is possible, however, to have $c_0 = 0$, at the expense of having a slightly more complicated analysis using a more general version of the Poincaré-Friedrichs inequality (cf. Example 11.1). Likewise, it is not strictly necessary to have smooth boundary conditions. Certain classes of square-integrable functions would do, but this would entail a more detailed knowledge of trace operators, which we omit here.